Grokking Data Structures & Algorithms for Coding Interviews

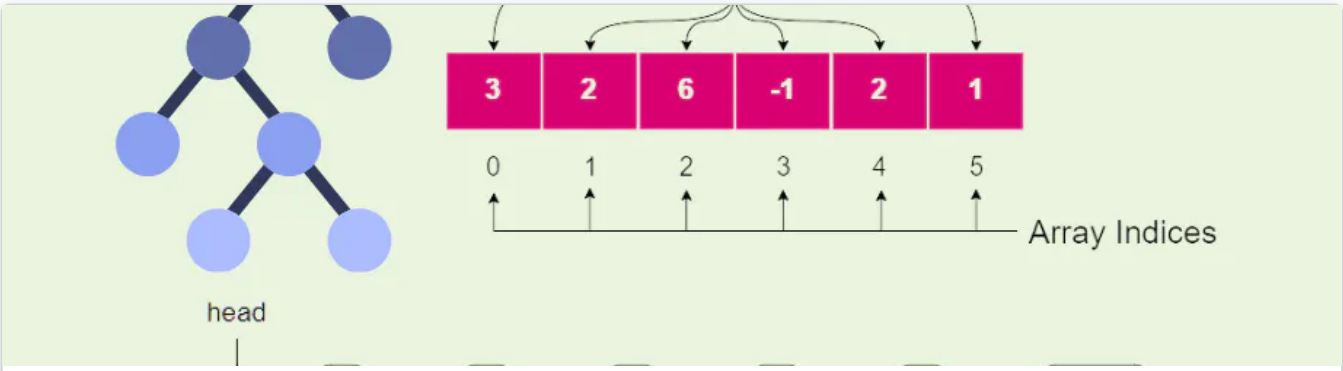
Unlock Coding Interview Success: Dive Deep into Data Structures and Algorithms!

Level: Beginner | Study Time: 50h | Lessons: 165 | Playgrounds : 185

Course Overview

In this comprehensive course, we delve deep into the world of data structures & algorithms, covering both the commonly used and the more advanced topics. Participants will learn the principles, usage, and implementation of various data structures, along with their time and space complexities. The course will feature multiple coding problems for each data structure and is tailored for coding interviews.

By the end of this course, learners will be adept at selecting and applying the most suitable data structure for any coding problem, ensuring optimized code performance. Prepare to elevate your coding skills and ace those coding interviews with confidence!



Start Reading

Certification on completing the course.

Download Certificate

Course Content

1. Introduction to Data Structures — 4 chapters

- Who Should Take This Course?
- Understanding Data Structures
- Types of Data Structures
- An Overview of Big-O

2. Array — 10 chapters

- Introduction to Arrays
- Arrays in Different Programming Languages
- Problem 1: Running Sum of 1d Array (easy)
- Solution: Running Sum of 1d Array (easy)
- Problem 2: Contains Duplicate (easy)
- Solution: Contains Duplicate

Solution: Left and Right Sum Differences (easy)

Problem 4: Find the Highest Altitude(easy)

Solution: Find the Highest Altitude (easy)

3. Matrix

7 chapters

Introduction to Matrix

Problem 1: Richest Customer Wealth (easy)

Solution: Richest Customer Wealth (easy)

Problem 2: Matrix Diagonal Sum (easy)

Solution: Matrix Diagonal Sum (easy)

Problem 3: Row With Maximum Ones(easy)

Solution: Row With Maximum Ones(easy)

4. Stack

22 chapters

Introduction to Stack

Operations on Stack

Implementing Stack Data Structure

Applications of Stack

Problem 1: Balanced Parentheses

Solution: Balanced Parentheses

Problem 2: Reverse a String

Solution: Reverse a String

Problem 3: Decimal to Binary Conversion

Problem 4: Next Greater Element

Solution: Next Greater Element

Problem 5: Sorting a Stack

Solution: Sorting a Stack

Problem 6: Simplify Path

Solution: Simplify Path

Problem 7: Remove All Adjacent Duplicates In String

Solution: Remove All Adjacent Duplicates In String

Problem 8: Removing Stars From a String

Solution: Removing Stars From a String

Problem 9: Make The String Great

Solution: Make The String Great

5. Queue

16 chapters

Introduction to Queues

Working with Simple Queues

Diving Deeper – Circular Queues and Deques

Applications and Advanced Concepts

Problem 1: Reverse a Queue

Solution: Reverse a Queue

Problem 2: Implement Stack using Queues

Solution: Implement Stack using Queues

Problem 3: Generate Binary Numbers from 1 to N

Problem 4: Palindrome Check using Queue

Solution: Palindrome Check using Queue

Problem 5: Zigzag Iterator

Solution: Zigzag Iterator

Problem 6: Max of All Subarrays of Size 'k'

Solution: Max of All Subarrays of Size 'k'

6. LinkedList

13 chapters

Introduction to LinkedList

Operations on Singly Linked List

Operations on Doubly Linked List

Problem 1: Reverse Linked List (easy)

Solution: Reverse Linked List (easy)

Problem 2: Remove Duplicates from Sorted List (easy)

Solution: Remove Duplicates from Sorted List (easy)

Problem 3: Merge Two Sorted Lists (easy)

Solution: Merge Two Sorted Lists (easy)

Problem 4: Find if Doubly Linked List is a Palindrome (easy)

Solution: Find if Doubly Linked List is a Palindrome (easy)

Problem 5: Swap Nodes in Pairs (medium)

Solution: Swap Nodes in Pairs (easy)

7. Tree & Binary Search Tree

16 chapters

Binary Search Tree (BST)

- Maximum Depth (or Height) of Binary Tree (easy)
- Solution: Maximum Depth (or Height) of Binary Tree (easy)
- Balanced Binary Tree (easy)
- Solution: Balanced Binary Tree (easy)
- Minimum Difference Between BST Nodes (easy)
- Solution: Minimum Difference Between BST Nodes (easy)
- Range Sum of BST (easy)
- Solution: Range Sum of BST (easy)
- Kth Smallest Element in a BST (medium)
- Solution: Kth Smallest Element in a BST (medium)
- Closest Binary Search Tree Value (medium)
- Solution: Closest Binary Search Tree Value (medium)
- Merge Two Binary Trees (medium)
- Solution: Merge Two Binary Trees (medium)

8. Hash Table (aka Hashmap or Dictionary)

13 chapters

- Introduction to Hashing
- Introduction to Hash Tables
- Issues with Hash Tables
- Problem 1: First Non-repeating Character (easy)
- Solution: First Non-repeating Character (easy)
- Problem 2: Largest Unique Number (easy)

Problem 3: Maximum Number of Balloons (easy)

Solution: Maximum Number of Balloons (easy)

Problem 4: Longest Palindrome(easy)

Solution: Longest Palindrome(easy)

Problem 5: Ransom Note (easy)

Solution: Ransom Note (easy)

9. HashSet

9 chapters

Introduction to HashSets

Problem 1: Counting Elements (easy)

Solution: Counting Elements (easy)

Problem 2: Jewels and Stones (easy)

Solution: Jewels and Stones (easy)

Problem 3: Unique Number of Occurrences (easy)

Solution: Unique Number of Occurrences (easy)

Problem 4: Longest Substring Without Repeating Characters (medium)

Solution: Longest Substring Without Repeating Characters (medium)

10. Heap

9 chapters

Introduction to Heap

Take Gifts From the Richest Pile(easy)

Solution: Take Gifts From the Richest Pile(easy)

Sort Characters By Frequency (easy)

Minimum Cost to Connect Sticks(medium)

Solution: Minimum Cost to Connect Sticks(medium)

Find the Median of a Number Stream (medium)

Solution: Find the Median of a Number Stream

11. Graph

11 chapters

Introduction to Graph

Graph Representations

Graph as an Abstract Data Type (ADT)

Graph Traversal - Depth First Search(DFS)

Graph Traversal - Breadth First Search (BFS)

Problem 1: Find if Path Exists in Graph(easy)

Solution: Find if Path Exists in Graph(easy)

Problem 2: Number of Provinces (medium)

Solution: Number of Provinces (medium)

Problem 3: Minimum Number of Vertices to Reach All Nodes(medium)

Solution: Minimum Number of Vertices to Reach All Nodes(medium)

12. Trie (aka Prefix Tree)

11 chapters

Introduction to Trie

Implement Trie (Prefix Tree) (medium)

Solution: Implement Trie (Prefix Tree)

Index Pairs of a String (easy)

- Extra Characters in a String (medium)
- Solution: Extra Characters in a String
- Search Suggestions System (medium)
- Solution: Search Suggestions System
- Design Add and Search Words Data Structure (medium)
- Solution: Design Add and Search Words Data Structure

13. Greedy Algorithm

13 chapters

- Introduction to Greedy Algorithm
- Valid Palindrome II (easy)
- Solution: Valid Palindrome II
- Maximum Length of Pair Chain (medium)
- Solution: Maximum Length of Pair Chain
- Minimum Add to Make Parentheses Valid (medium)
- Solution: Minimum Add to Make Parentheses Valid
- Remove Duplicate Letters (medium)
- Solution: Remove Duplicate Letters
- Largest Palindromic Number (Medium)
- Solution: Largest Palindromic Number
- Removing Minimum and Maximum From Array (medium)
- Solution: Removing Minimum and Maximum From Array

14. Divide and Conquer Algorithm

11 chapters

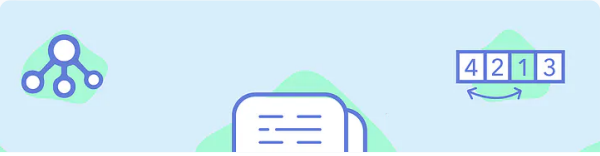
- Majority Element (easy)
- Solution: Majority Element
- Longest Nice Substring (easy)
- Solution: Longest Nice Substring
- Sort List (medium)
- Solution: Sort List
- Beautiful Array (medium)
- Solution: Beautiful Array
- Median of Two Sorted Arrays (medium)
- Solution: Median of Two Sorted Arrays

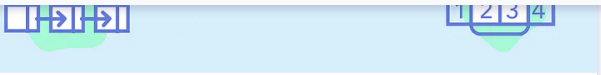
Related Courses



Grokking Dynamic Programming Patterns for Coding Interviews

Start Reading





Grokking the Coding Interview: Patterns for Coding Questions

Start Reading

What people say about our courses

Copyright © 2024 Designgurus, Inc. All rights reserved.